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MID TERM EXAMINATIONS – March 2026

Programme	: **B.Tech.**	Semester	:	**Winter 2025-26**
Course Title	: **Electric Circuits and Systems**	Course Code	:	**EEE1001**
Date/ Session	: **16th March 2026 / Session-II**	Slot	:	**A14+D11+D12**
Time	: **1 ½ hours**	Max. Marks	:	**50**

Answer all the Questions

Q.No.	Sub. Sec.	Question Description	Marks
1	(a)	Find the source current of the network shown in Fig. 1 using the Star-delta transformation.	5

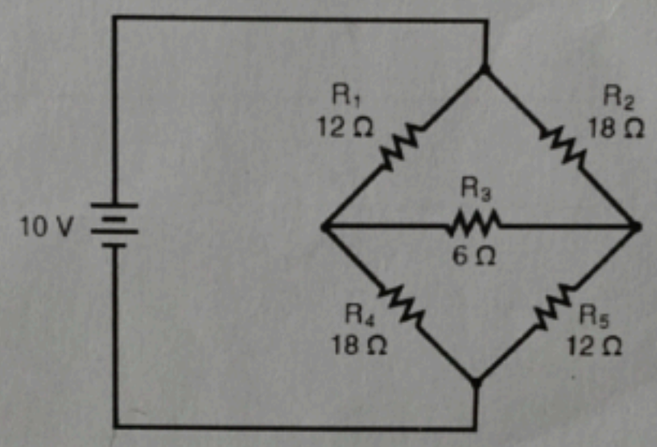


Fig. 1

1	(b)	Determine V_0 in the circuit given in Fig. 2.	5

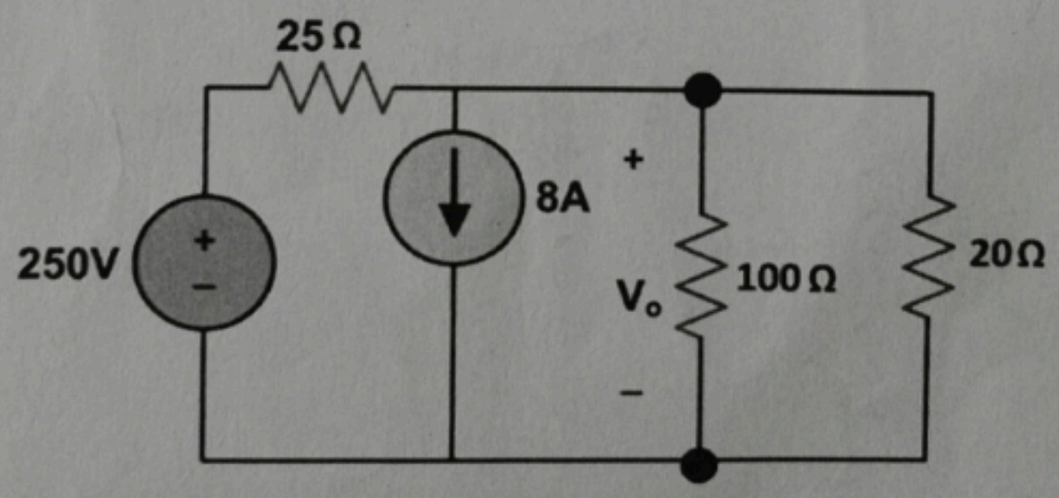


Fig. 2

2		Transform the circuit given in Fig. 3 into Norton equivalent Circuit across terminal a-b and determine the current across the load resistance taking R_L= 6 ohm.	10

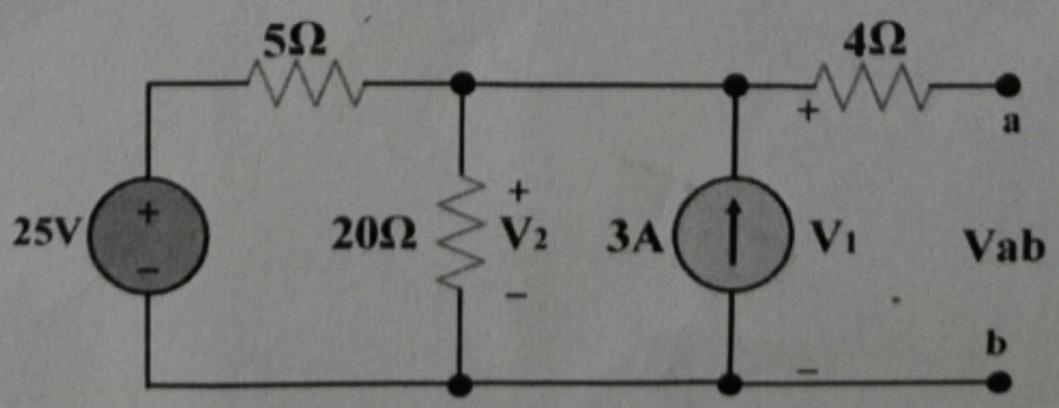
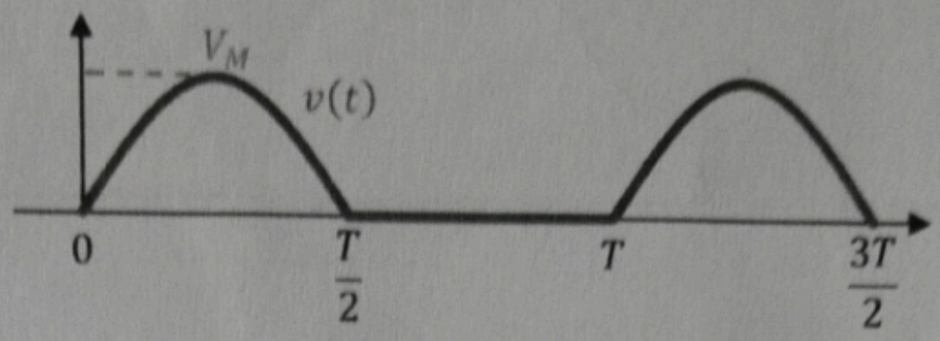
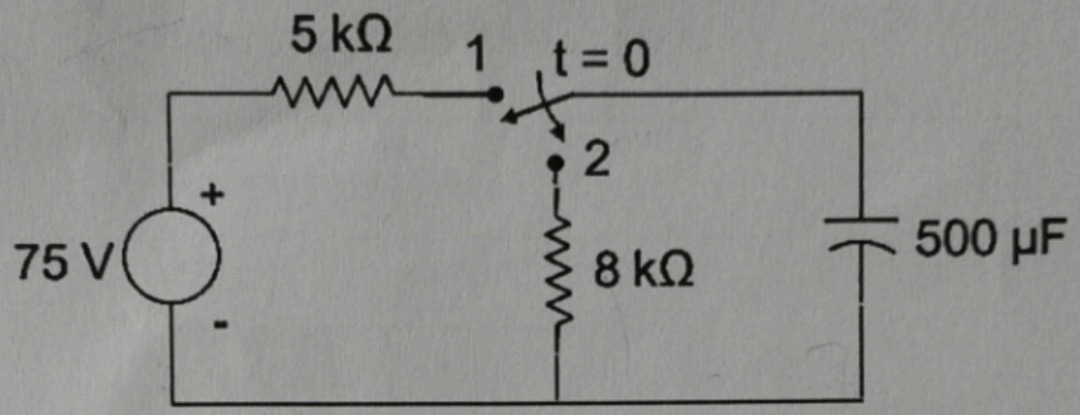


Fig. 3

3	(a)	Calculate the RMS value and Average value of the sinusoidal waveform shown in Fig. 4  Fig. 4	5
3	(b)	A coil of resistance 10Ω and inductance of 0.1 H is connected in series with a capacitance of 150μF across 200 V, 50 Hz supply. Calculate impedance, current and power factor consumed.	5
4		The switch in the circuit shown was in open position for a long time. It is closed at time t = 0. Find ic(t) for time t > 0.  Fig. 5	10
5		A coil of 100 turns is wound uniformly over an insulator ring with a mean circumference of 2m and a uniform sectional area of 0.025cm2. If the coil is carrying a current of 2A. Calculate (a) the mmf of the circuit, (b) the magnetic field intensity, (c) flux density and (d) total flux.	10

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